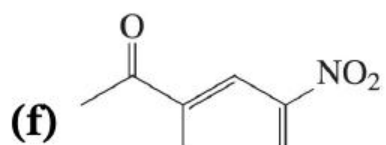
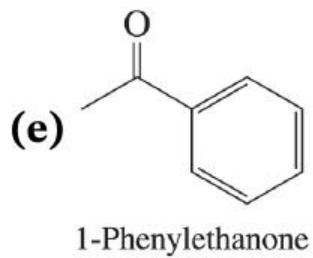
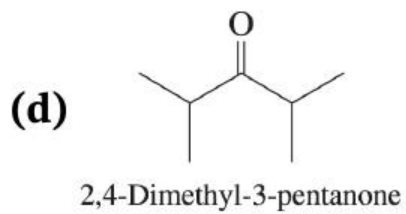
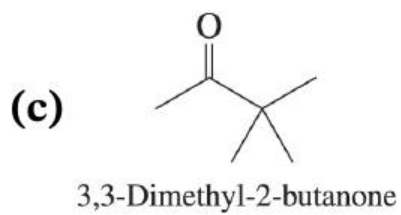
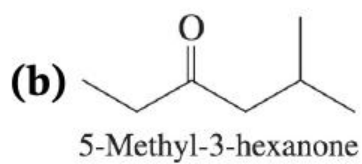
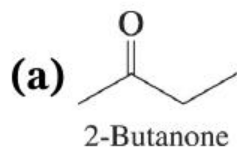
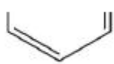


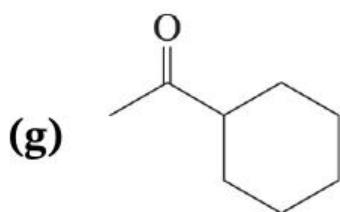
Solutions to Problems

27.





1-(3-Nitrophenyl)ethanone



1-Cyclohexylethanone

28.

(a) 2,4-Dimethyl-3-pentanone

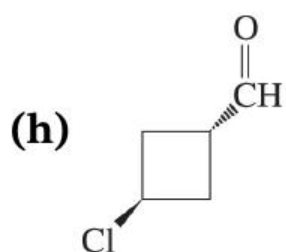
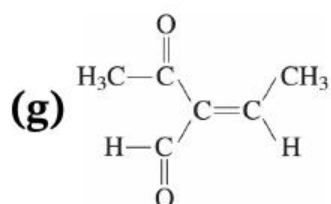
(b) 4-Methyl-3-phenylpentanal

(c) 3-Buten-2-one

(d) *trans*-4-Chloro-3-butenal

(e) 4-Bromo-2-cyclopentenone

(f) *cis*-2-Ethanoyl-3-phenylcyclohexanone (*cis*-2-acetyl-3-phenylcyclohexanone)



29. Degree of unsaturation for

$C_8H_{12}O$, $H_{sat} = 16 + 2 = 18$; degree of unsaturation $= \frac{(18 - 12)}{2} = 3$ π

$C_8H_{12}O$, $H_{sat} = 16 + 2 = 18$; degree of unsaturation $= \frac{(18 - 12)}{2} = 3$ π

bonds or rings present.

(a) ^{13}C : ^{13}C : Molecule contains $\text{C}=\text{O}$ ($\delta = 199.2$) and

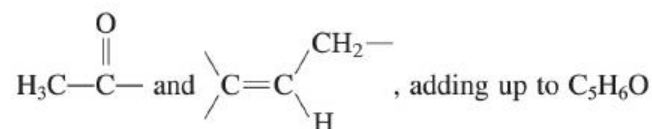
$\text{C}=\text{C}$ ($\delta = 139.7$ and 140.9).

^1H : ^1H : Obvious features include $\delta=2.15$ (s, 3 H),

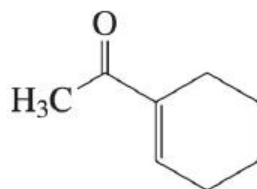
$\delta = 2.15$ (s, 3 H), for $\text{CH}_3\text{C}(=\text{O})-$, and $\delta=6.78$ (t, 1 H),

$\delta = 6.78$ (t, 1 H), for CH_2- attached to a $\text{C}=\text{C}$ group. Note the absence of any

other alkene hydrogens. So we can start with these pieces:



Still needed are three more C's, six more H's, and another degree of unsaturation, which will have to be a ring. A simple way to put together a trial structure would be to finish a six-membered ring with three CH_2 groups, and then attach the acetyl group:



This is not the only possible answer, but it is the actual molecule that gives the indicated spectra.

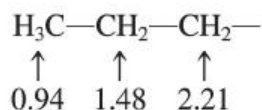
(b) ^{13}C NMR ^{13}C NMR spectrum: One $\text{C}=\text{O}$ ($\delta=193.2$)

and **two** $\text{C}=\text{C}$ groups

($\delta=129.0, 135.2, 146.7$, and $\delta=129.0, 135.2, 146.7$, and 152.5) this time.

^1H NMR ^1H NMR spectrum: The carbonyl group is an

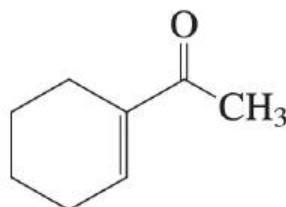
aldehyde ($\delta = 9.56$ for $\text{—}\overset{\text{O}}{\parallel}\text{C—H}$). At the other end, we have



This adds up to $\text{C}_4\text{H}_8\text{O}$, $\text{C}_4\text{H}_8\text{O}$, leaving C_4H_4 to account for. All four of these H's are alkene hydrogens $\delta = 5.8\text{--}7.1$, so these could most simply be two —CH=CH— groups. The result:



30. Each is a conjugated carbonyl compound, giving an intense UV absorption with $\lambda_{\text{max}} > 200 \text{ nm}$. The first spectrum matches

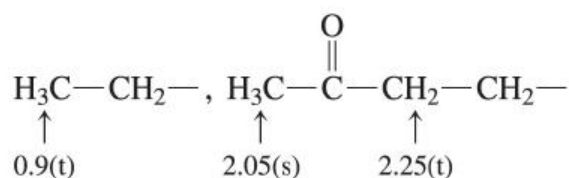


with $\pi \rightarrow \pi^*$ absorption at 232 nm , and carbonyl $n \rightarrow \pi^*$ absorption at 308 nm . The second spectrum matches the diene-aldehyde, with the longer wavelength 272 nm band corresponding to the $\pi \rightarrow \pi^*$ absorption of the more extended conjugated system.

31. MS: M^+ of 128 confirms that $\text{C}_8\text{H}_{16}\text{O}$ is the molecular formula as well

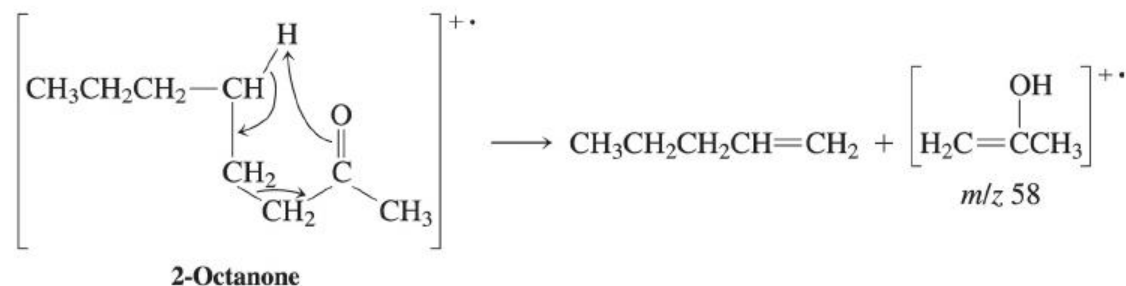
$$H_{\text{sat}} = 16 + 2 = 18; \text{ degree of unsaturation} = \frac{(18 - 16)}{2} = 1 \pi \text{ bond or ring present}$$

IR, UV spectra: A ketone $\text{C}=\text{O}$ appears to be present
NMR spectrum:



are likely pieces, adding up to $\text{C}_6\text{H}_{12}\text{O}$; $\text{C}_6\text{H}_{12}\text{O}$; only C_2H_4 C_2H_4 are left to add in. Is 2-octanone a reasonable answer?

MS: Base peak (m/z 43) (m/z 43) is $\left[\text{CH}_3\text{C} \begin{array}{c} \text{O} \\ \parallel \end{array} \right]^+$; next largest is m/z 58, m/z 58, consistent with *McLafferty rearrangement* as follows:



This answer seems quite reasonable.

- 32.** Be sure you've read the problem correctly. The given structures are your *starting materials*, and you are asked to identify a reagent that will turn each one of them *into* 3-hexanone.

(a) PCC, CH₂Cl₂ or CrO₃, H₂SO₄; PCC, CH₂Cl₂ or CrO₃, H₂SO₄,

(b) H₂O, H⁺; H₂O, H⁺;

(c) HgCl₂, CaCO₃, CH₃CN, H₂O; HgCl₂, CaCO₃, CH₃CN, H₂O;

(d) H₂O, H⁺ or ⁻OH; H₂O, H⁺ or ⁻OH;

(e) 1. O₃; O₃; 2. (CH₃)₂S or Zn, CH₃COOH; (CH₃)₂S or Zn, CH₃COOH;

(f) H₂O, H⁺ or ⁻OH; H₂O, H⁺ or ⁻OH;

(g) Hg²⁺, H₂O or 1. R₂BH (R = cyclohexyl),
Hg²⁺, H₂O or 1. R₂BH (R = cyclohexyl), 2. H₂O₂, H₂O, ⁻OH;
H₂O₂, H₂O, ⁻OH;

(h) H₂O, ⁻OH. H₂O, ⁻OH.

33.

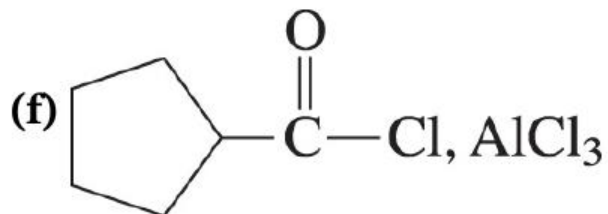
(a) CrO_3 , H_2SO_4 , acetone, or MnO_2 , CH_2Cl_2 (better);
 CrO_3 , H_2SO_4 , acetone, or MnO_2 , CH_2Cl_2 (better);

(b) PCC, CH_2Cl_2 ; PCC, CH_2Cl_2 ;

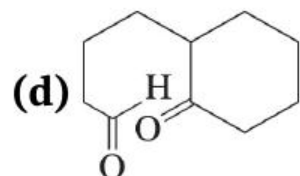
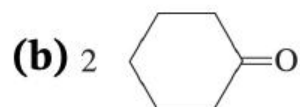
(c) 1. O_3 , CH_2Cl_2 , 2. Zn , CH_3COOH , H_2O ; 1. O_3 , CH_2Cl_2 , 2. Zn , CH_3COOH , H_2O ;

(d) HgSO_4 , H_2O , H_2SO_4 ; HgSO_4 , H_2O , H_2SO_4 ;

(e) same as (d);



34.



35.

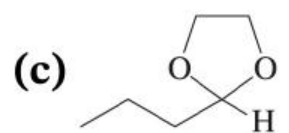
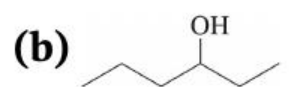
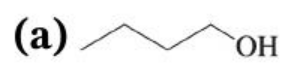
(a) Ask “How electrophilic is the carbon in question?”:

$(\text{CH}_3)_2\text{C}=\text{O} + \text{H} > (\text{CH}_3)_2\text{C}=\text{O} > (\text{CH}_3)_2\text{C}=\text{NH}$ $(\text{CH}_3)_3\text{COH} > (\text{CH}_3)_3\text{CO} > (\text{CH}_3)_3\text{CK}$
by electronegativity.

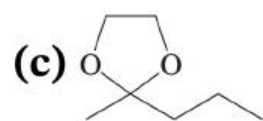
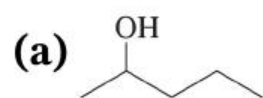
(b) $\overset{\text{O}}{\parallel}\overset{\text{O}}{\parallel}\overset{\text{O}}{\parallel}\text{CH}_3\text{CCCCH}_3 > \overset{\text{O}}{\parallel}\overset{\text{O}}{\parallel}\text{CH}_3\text{CCCH}_3 > \overset{\text{O}}{\parallel}\text{CH}_3\text{CCH}_3$ Adjacent carbonyl groups increase reactivity by reinforcing the $\delta^+\delta^+$ character of their respective carbons.

(c) $\text{BrCH}_2\text{CHO} > \text{CH}_3\text{CHO} > \text{BrCH}_2\text{COCH}_3 > \text{CH}_3\text{COCH}_3$
 $\text{BrCH}_2\text{CHO} > \text{CH}_3\text{CHO} > \text{BrCH}_2\text{COCH}_3 > \text{CH}_3\text{COCH}_3$ Aldehydes are more reactive than ketones and halogen substituents increase reactivity.

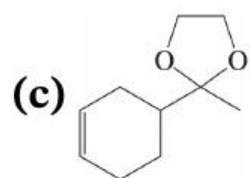
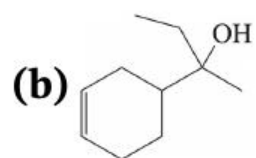
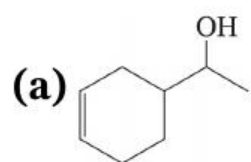
36.



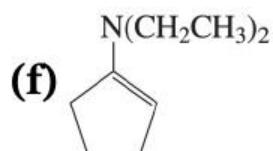
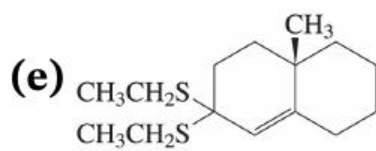
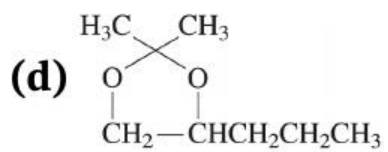
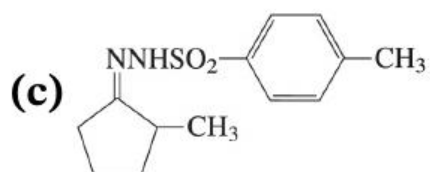
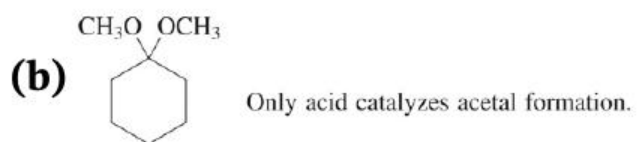
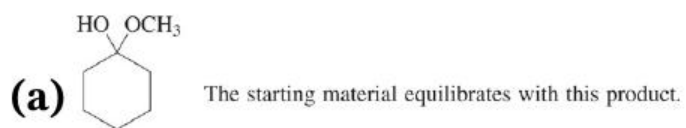
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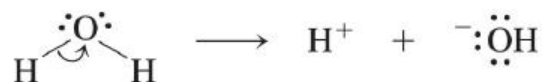
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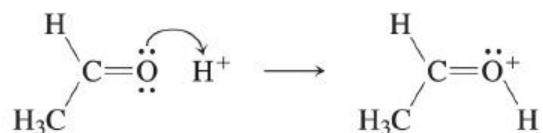
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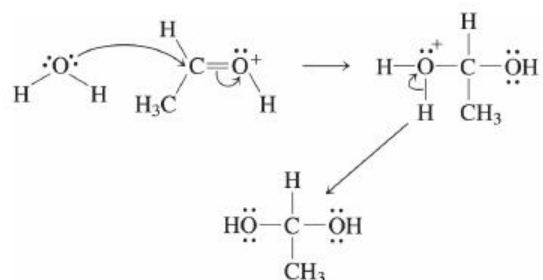
10. (1) Just an acid dissociating. The arrow goes the wrong way. Don't point the arrow towards the H if you are making $H^+ + OH^-$. Here's the right answer:



- (2) Now it is the reverse: the combination of an acidic proton with a base. Again, the arrow is backwards. There are no electrons in H^+ , so an arrow can't start there. Arrows move electrons, not atoms. So:

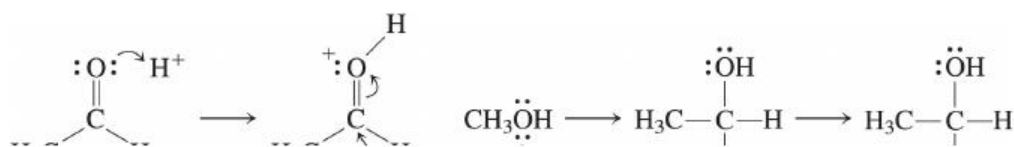


- (3) Better. At least the arrows go the right way. However, under acidic conditions hydroxide doesn't exist to any realistic degree. It is negative, and in acid only neutral or positive species should be present. Change it to water, then add the electrons, and complete the picture with proton loss:



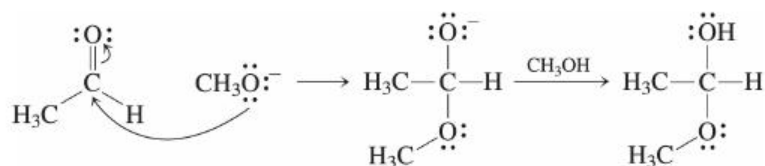
11.

(a) In acid:

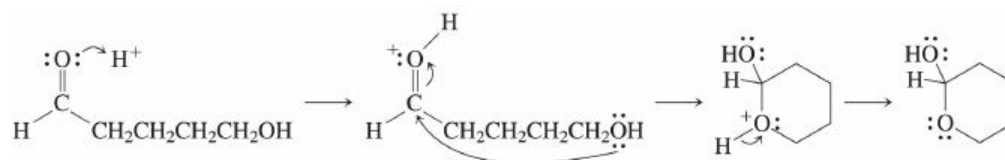




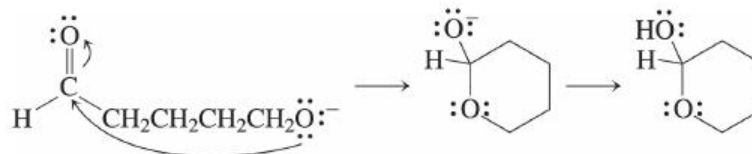
In base:



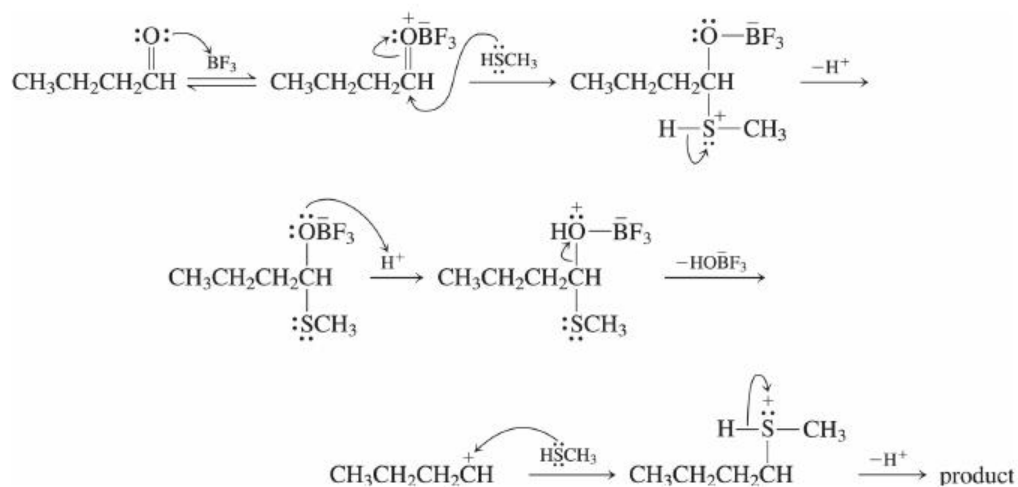
(b) In acid:



In base:

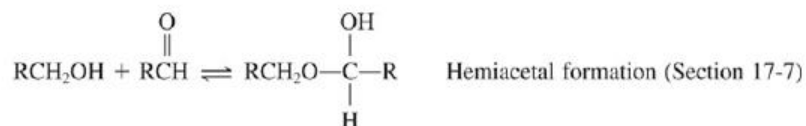


(c)

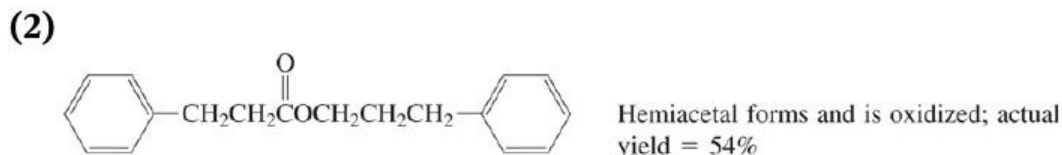
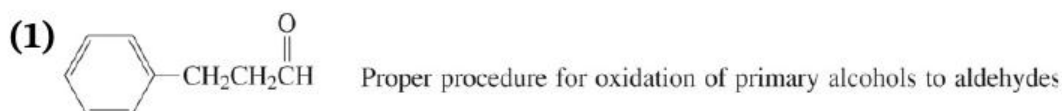




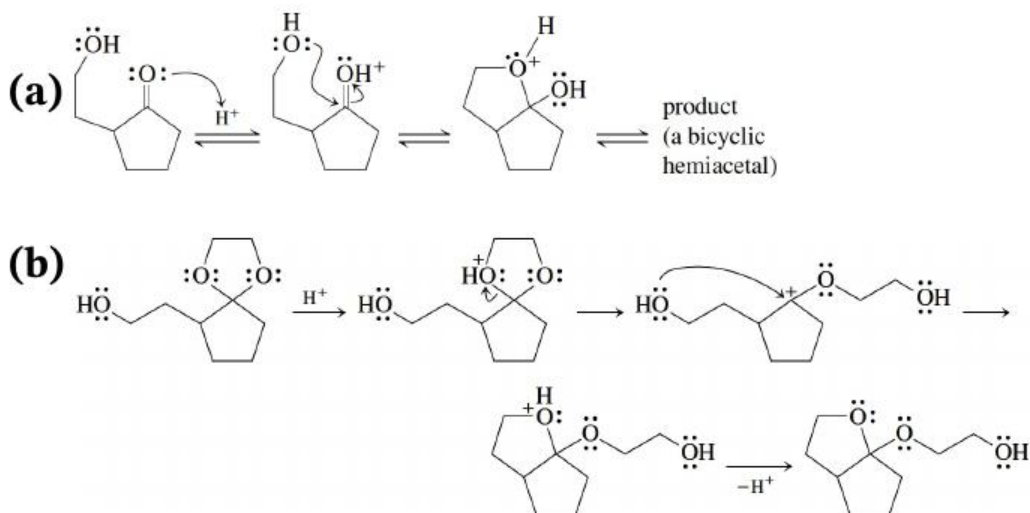
12. (a) **Ketone** hydrates do **not** contain the $\text{H}-\text{C}-\text{OH}$ unit that is required for further oxidation to occur. Oxidation beyond the ketone stage would require cleavage of a carbon-carbon bond, a difficult process (compare the Baeyer-Villiger oxidation; [Section 17-13](#)).
- (b) Think about it: If CrO_3 is added to an alcohol, what will be present in the mixture? There will be some aldehyde formed from oxidation, in the presence of an excess of **unreacted alcohol**. What reaction occurs between aldehydes and alcohols?



(c)



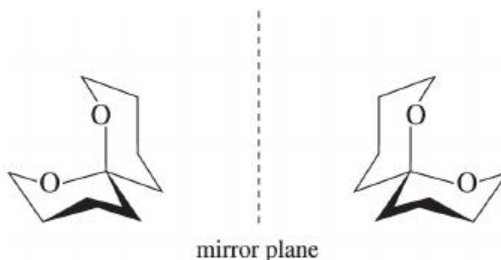
13.



(c) See text [sections 17-6 and 17-7](#) and [Study Guide p. 341](#).

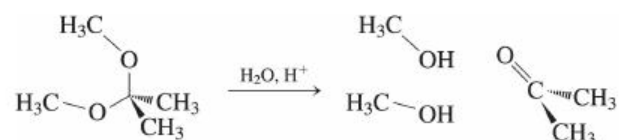
14.

(a) The structures are presented in a way that should remind you of one particular isomeric relationship. Imagine a vertical mirror placed between them. Do you see each as the reflection of the other? These are enantiomers.

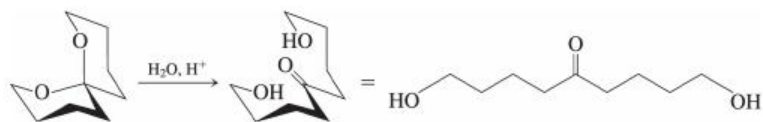


(b) Both molecules contain acetal functions: two oxygen atoms attached by single bonds to a common carbon. (Neither is an OHOH group; otherwise we'd be looking at hemiacetals.)

(c) The acetal function undergoes hydrolysis under aqueous acidic conditions to a carbonyl compound. The common carbon referred to above becomes double bonded to oxygen. The two oxygens are released as hydroxy functions. First, consider a simpler example:

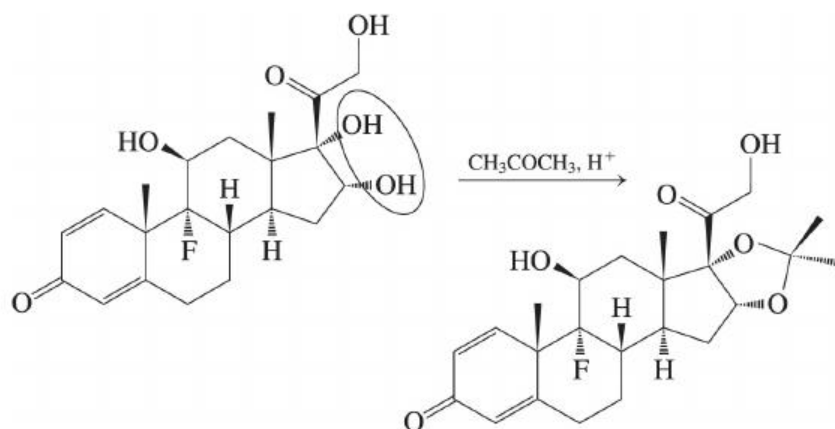


We drew this example in just an orientation to make clearer its resemblance to the molecule in the problem. Now take our molecule and proceed similarly:

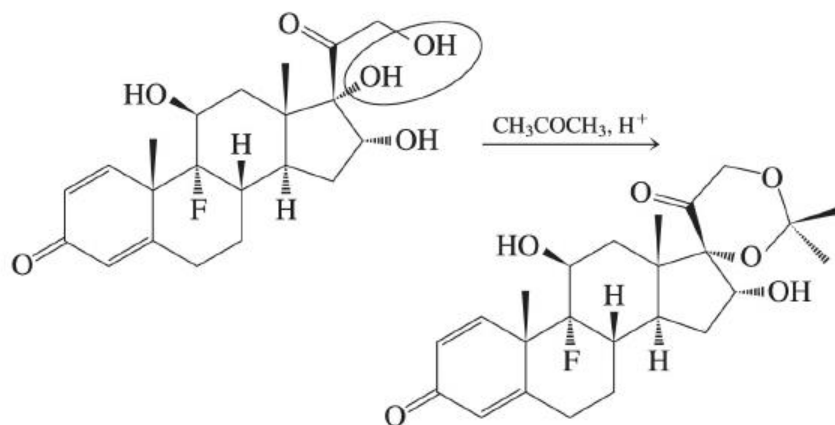


Do you see how these two examples parallel each other? Together they make a good exercise in pattern recognition, a very valuable tool in solving reaction problems. Note that the final product is achiral; both enantiomers of the starting compound hydrolyze to give this same molecule.

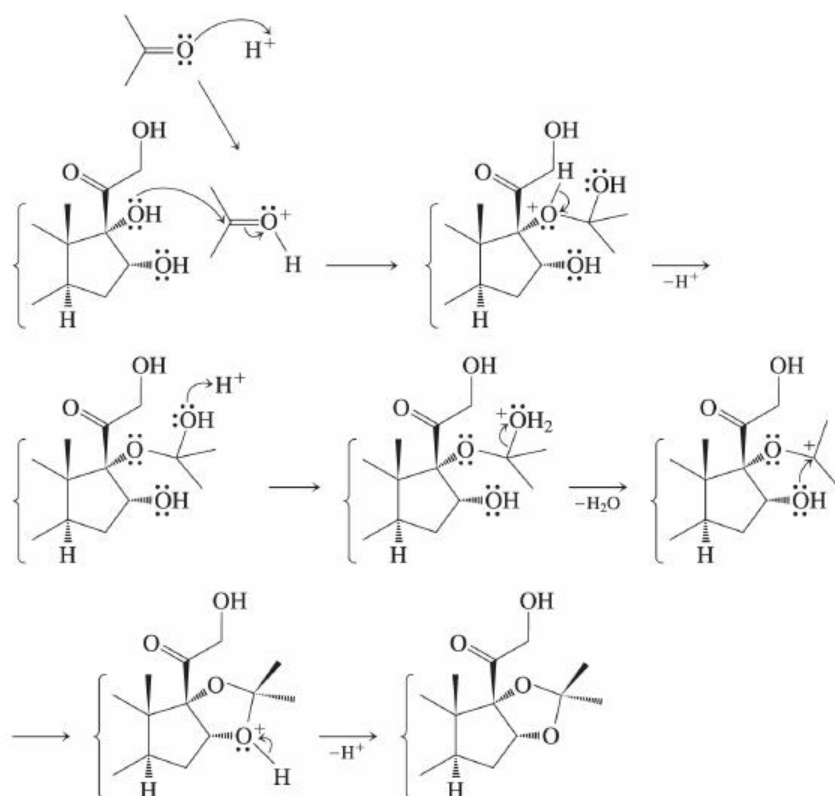
15. Most sensible would be cyclic acetal formation using two alcohol groups in close proximity. For example, we can make a five-membered ring:



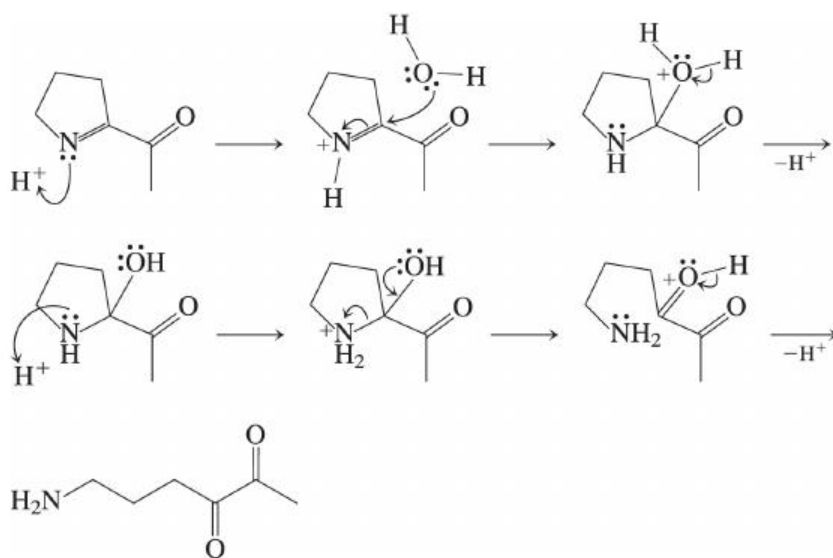
Or, we can make a six-membered ring:



The mechanisms that form each possible product are the same. To save space let's abbreviate the steroid structure and just show the formation of the first one:



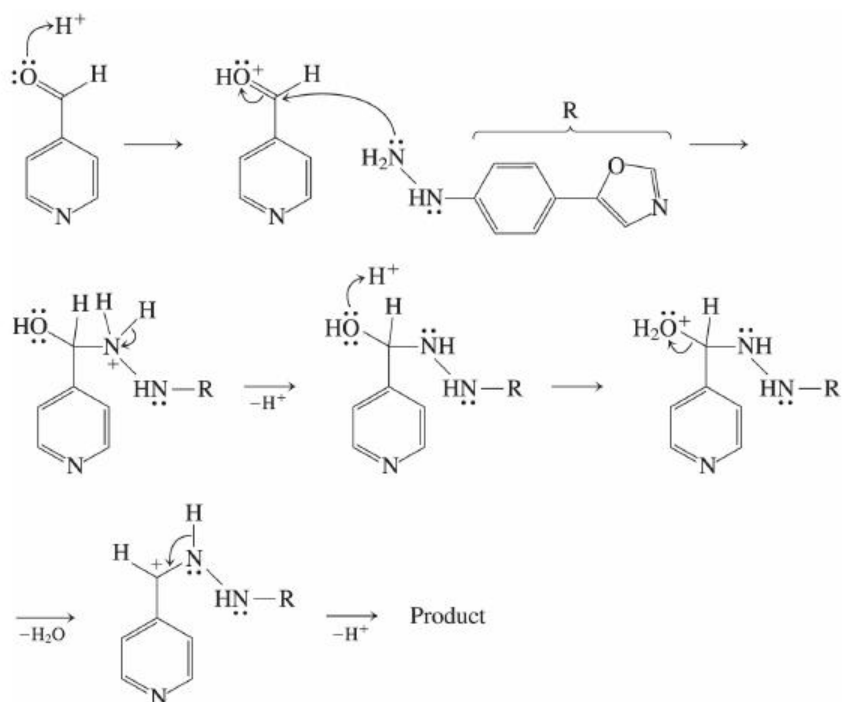
16. The $\text{C}=\text{N}^{\text{CaN}}$ double bond in the ring is an imine function and is hydrolyzable.



Under the acidic conditions of the hydrolysis, the terminal nitrogen

will be protonated. Neutralization of the acid will give the free amino function.

17. "Compound B" is a hydrazone made from the phenylhydrazine and aromatic aldehyde derivatives shown.



18.

(a) NaBH_4 ; LiAlH_4 , 2. H_2O , H^+ ;

(b) $\text{HSCH}_2\text{CH}_2\text{SH}$, ZnCl_2 ;

(c) 1. $\text{CH}_3\text{CH}_2\text{MgCl}$ (or $\text{CH}_3\text{CH}_2\text{Li}$), 2. H_2O , H^+ ;

(d) NaCN , H^+ ;

(e) RCO_3H ;

(f) $(\text{CH}_3\text{CH}_2)_2\text{NH}$, H^+ ;

(g) $(\text{C}_6\text{H}_5)_3\text{P}=\text{CH}_2$;

(h) Zn(Hg) , HCl , Δ , or H_2NNH_2 , H_2O , $-\text{OH}$, Δ ;

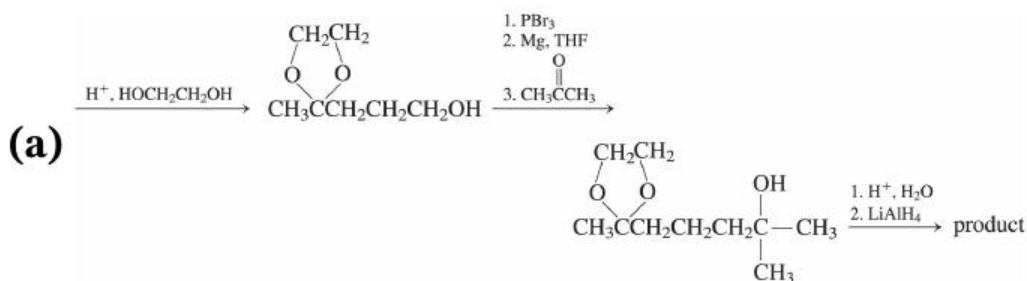
(i) CH_3OH , H^+ ;

(j) $(\text{C}_6\text{H}_5)_3\text{P}=\text{CHCH}_2\text{CH}_2\text{CH}_3$;

(k) 1. $\text{C}_6\text{H}_5\text{MgCl}$ (or $\text{C}_6\text{H}_5\text{Li}$), 2. H_2O ,

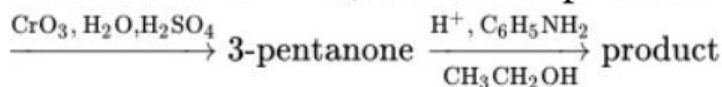
(l) $\text{C}_6\text{H}_5\text{NH}_2$, H^+ .

19.

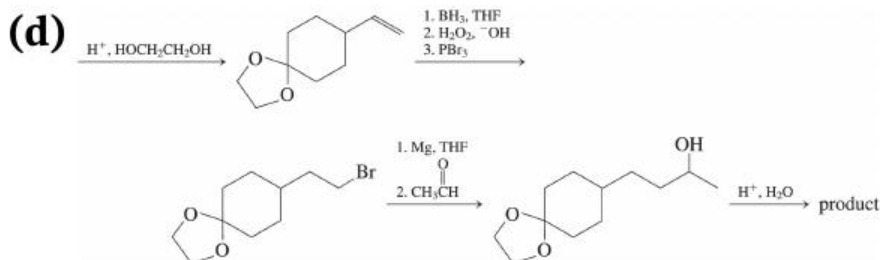
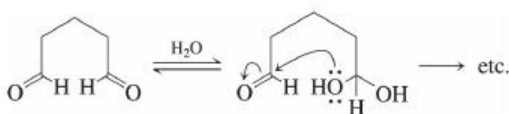


(b) $\rightarrow \text{CrO}_3, \text{H}_2\text{O}, \text{H}_2\text{SO}_4$ -pentanone

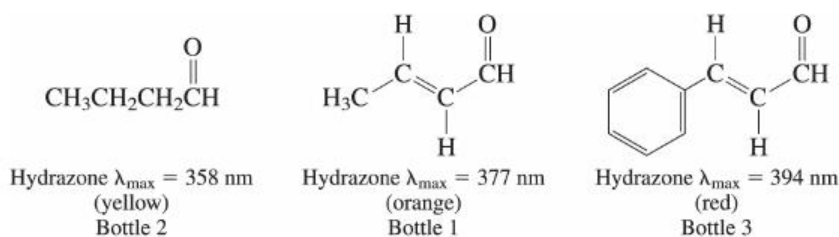
$\rightarrow \text{CH}_3\text{CH}_2\text{OHH}^+, \text{C}_6\text{H}_5\text{NH}_2$ product



(c) $\rightarrow 2\text{PCC}, \text{CH}_2\text{Cl}_2$ 1,5-pentanedial $\rightarrow \text{H}_2\text{O}$ product, via



50. Extended conjugation would be expected to shift the absorptions of these hydrazones to longer wavelengths, just as it does with the parent ketones. So,



51.

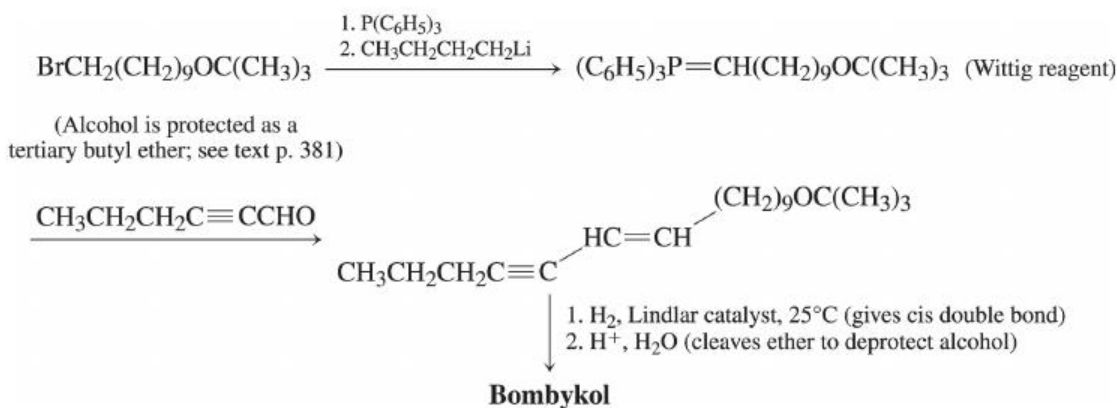
(a) H_2NNH_2 , H_2O , HO^- , Δ , H_2NNH_2 , H_2O , HO^- , Δ (Wolff-Kishner reduction of both carbonyl groups)

(b) H_2 , Pd-C, $\text{CH}_3\text{CH}_2\text{OH}$, H_2 , Pd-C, $\text{CH}_3\text{CH}_2\text{OH}$ (Selective reduction of alkene)

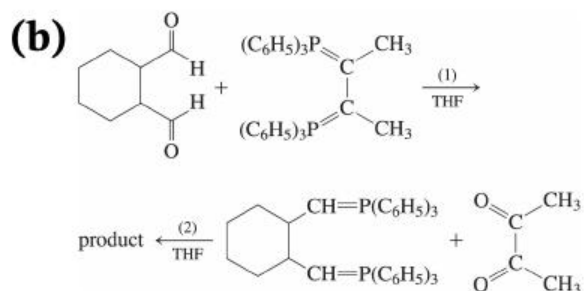
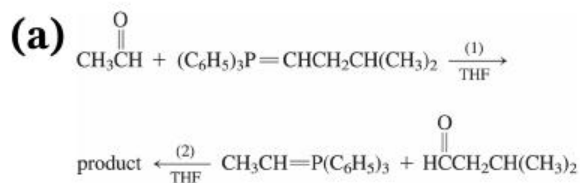
(c) LiAlH_4 , $(\text{CH}_3\text{CH}_2)_2\text{O}$, LiAlH_4 , $(\text{CH}_3\text{CH}_2)_2\text{O}$ (Selective reduction of aldehyde)

(d) H^+ , H^+ , cycloheptanone (Formation of an unusual acetal, nothing more)

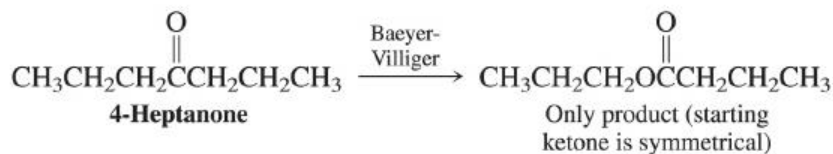
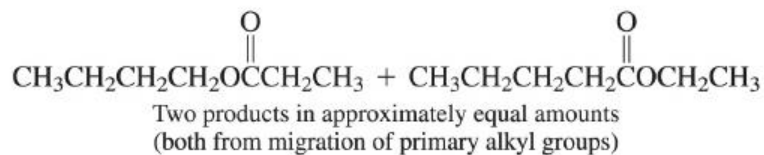
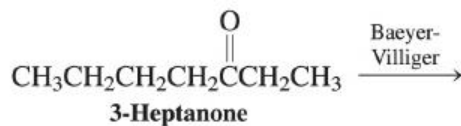
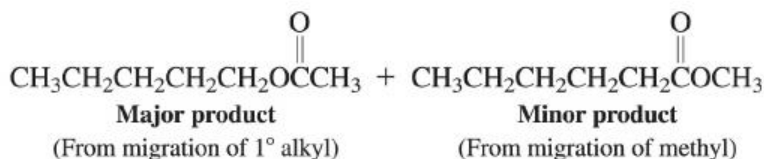
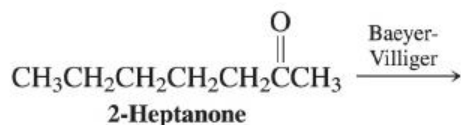
52. The alcohol group must be protected. The cis double bond derives from the triple bond.



53.



54. There are only three possible ketones: 2-heptanone, 3-heptanone, and 4-heptanone. So this is really a matching problem.



Compound A must be 4-heptanone; B is 2-heptanone; C is 3-heptanone.

55.

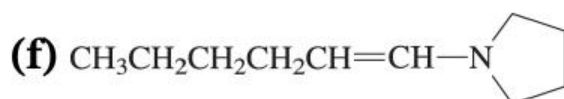


(b) 1-Hexanol

(c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}=\text{NOH}$

(d) Hexane

(e) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}(\text{CH}_3)_2$
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}(\text{CH}_3)_2$ (*E* and *Z*)

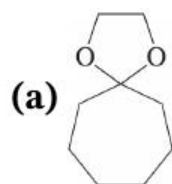


(g) Hexanoic acid + Ag metal

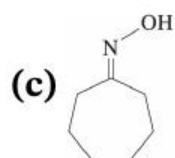
(h) Hexanoic acid

(i) 2-Hydroxyheptanenitrile $\left(\begin{array}{c} \text{O} \\ \parallel \\ \text{RCH} \end{array} \rightarrow \begin{array}{c} \text{OH} \\ | \\ \text{RCHCN} \end{array} \right)$

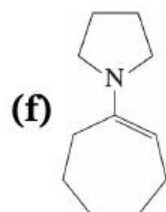
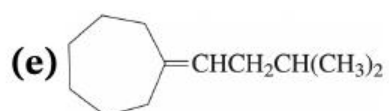
56.



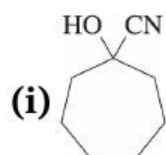
(b) Cycloheptanol



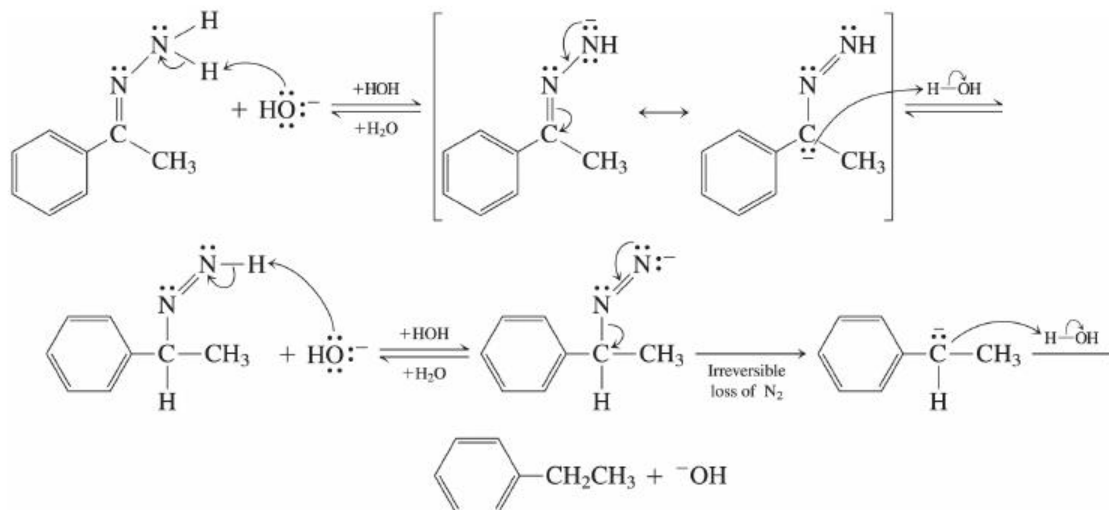
(d) Cycloheptane



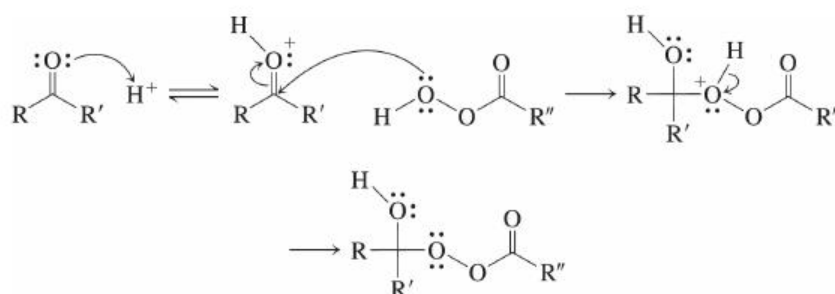
(g) and **(h)** No reaction. These are reactions of aldehydes only.



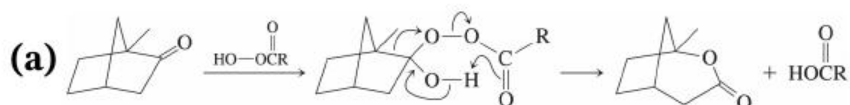
57.



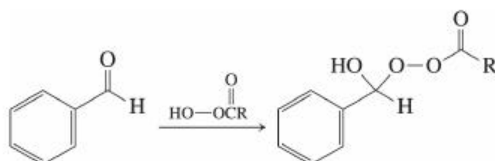
58. The reaction occurs under acidic conditions. Therefore:



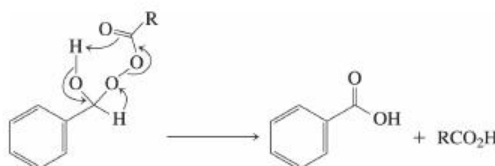
59.

See solution to [Problem 58](#)

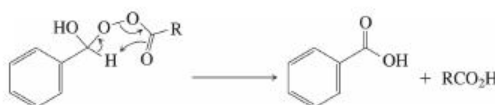
- (b) Begin by writing the product of addition of the peracid to the aldehyde, analogous to the first step in the Baeyer-Villiger mechanism with a ketone. We use benzaldehyde to illustrate.



This intermediate addition product can react via the cyclic mechanism already shown for ketones to give the observed carboxylic acid product. **H** is the migrating group:

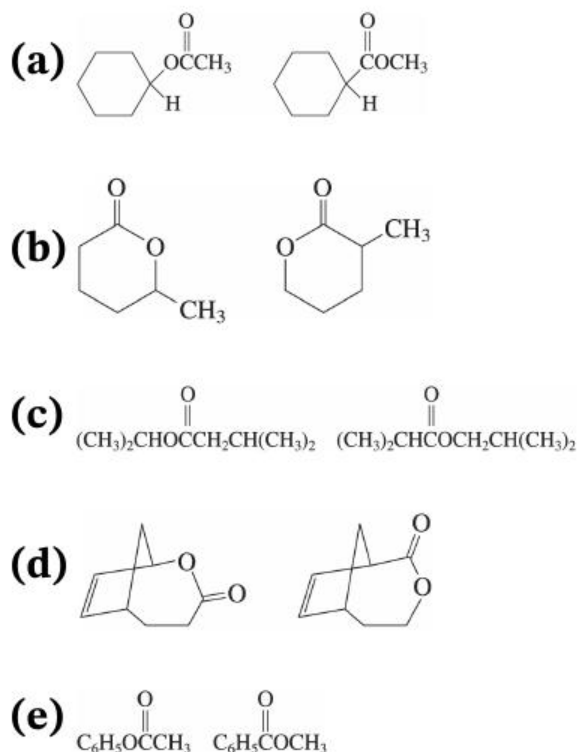


Alternatively, and more simply, an **E2-type** elimination could give the same end result. We show it here with the carbonyl oxygen acting as an internal base to remove the proton:

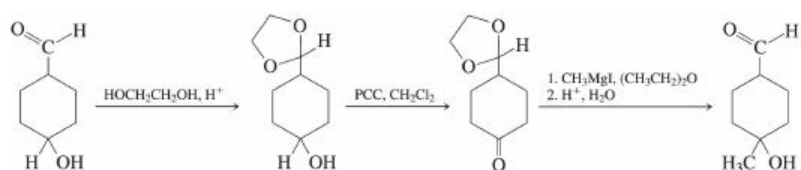


This second mechanism is an attractive option because (a) it is simpler, and (b) hydrogen normally displays low migratory aptitude. The latter observation is not consistent with the fact that the Baeyer-Villiger reaction of aldehydes with peracids generally leads to carboxylic acids.

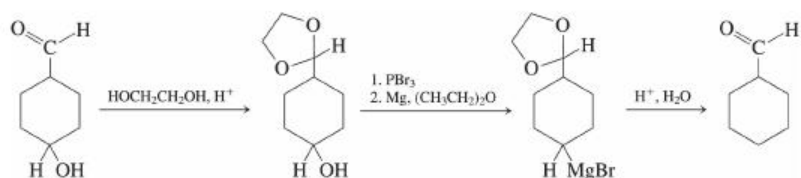
30. In each case insert **OO** on either side of the carbonyl **C**. Then refer to the information on migratory aptitudes in Section 17-13 to help you choose the preferred product. In the answers below, the **first** structure is the preferred one.



31. (a) Approach: You need a reaction between CH_3MgI and a ketone carbonyl group, so the aldehyde will have to be protected.



(b) Careful! Better protect the aldehyde again, or it will die when you apply any of several methods to get rid of the alcohol. Here's one:



Now go to work on the rest.

OH

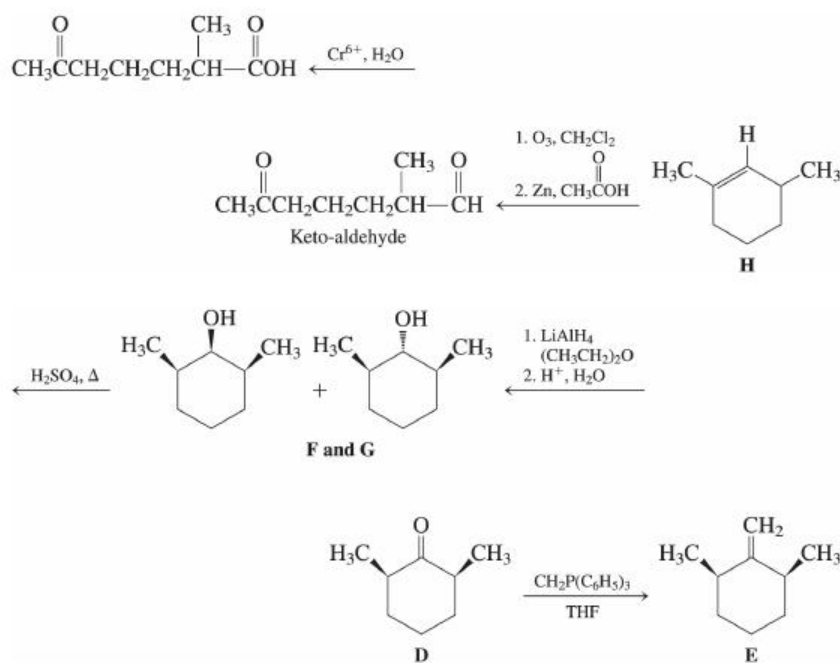
52. Here is the key issue: Cyclopropanone has more bond angle strain than does the corresponding hemiacetal. Why? The carbonyl carbon wants to be sp^2 hybridized and have 120° angles. Being stuck in the triangular ring with an angle near 60° , it experiences the strain of a bond angle compression of $120^\circ - 60^\circ = 60^\circ$. In contrast, the hemiacetal carbon wants only to be tetrahedral, with 109° bond angles. The strain it experiences is therefore less, corresponding to a compression of $109^\circ - 60^\circ = 49^\circ$ relative to the desired angle. We've seen situations like this before, in

the relative unreactivity of cyclopropane in radical halogenation ([Chapter 4, Problem 26](#)), and in the sluggishness of S_N2 displacements on halocyclopropanes ([Chapter 6, Problem 62](#)).

33. Below pH 2, the NN in NH_2OH is protonated $+\text{NH}_3\text{OH}$, so the nucleophilic atom is effectively gone. At pH 4, the NN is mostly free, but the solution is still acidic enough for some carbonyl groups to be made more electrophilic by protonation:

$\text{C}=\text{OH}^+$. Above pH 7, no carbonyl groups are protonated, so the rate is reduced to that of free NH_2OH attacking unactivated $\text{C}=\text{O}$ groups.

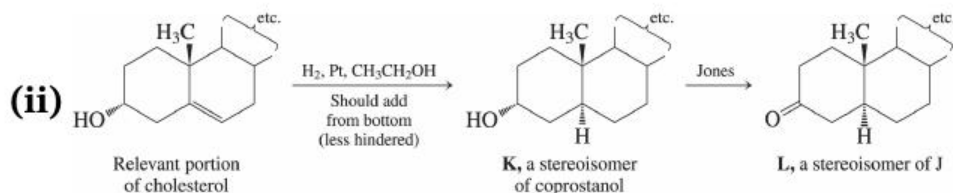
34. Work backward from given structural information.



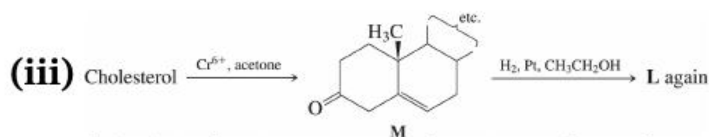
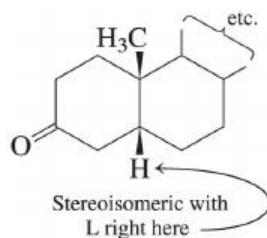
Note: F and G must be **secondary** alcohols because they have been formed by LiAlH_4 reduction of something (here, a ketone). The methyls in D must be cis. If they were trans, only one LiAlH_4 reduction product would be obtained.

35. The conclusions that may be reached from each piece of information are given to you.

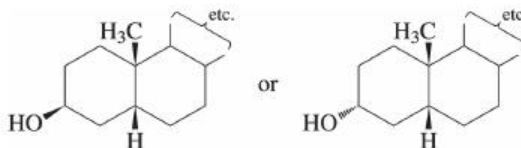
(i) Coprostanol is an alcohol, and J is a ketone.



It appears likely, then, that J is



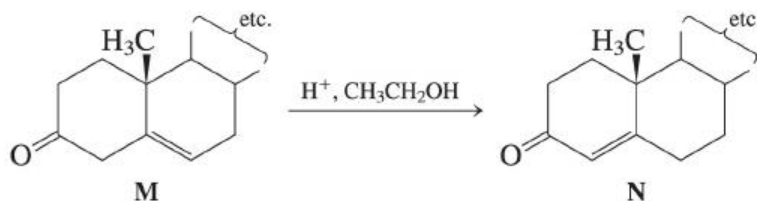
This is what you would expect, based on the information in (ii). So, what's coprostanol? Because it gives J on oxidation with Jones reagent, it must be one of the two alcohols



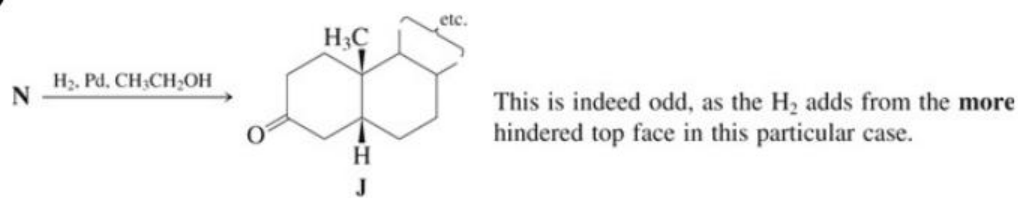
In fact, both are known and are labeled 3β - and 3α -coprostanol, respectively.

36.

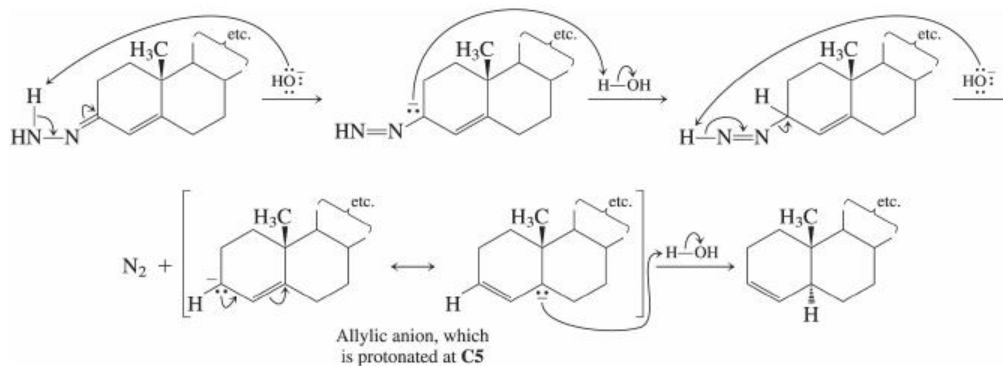
- (a) The UVUV spectrum indicates a **conjugated** ketone (compare UVUV data of compounds in Exercise 17-4 and of 3-buten-2-one in Section 17-3). So NN would appear to result from



(b)

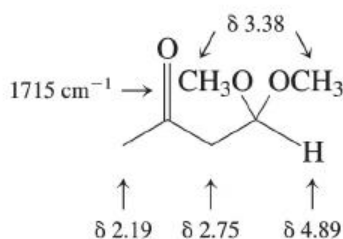


- (c) The hydrazone forms in the usual way ([Section 17-9](#)). Then



37. The main issue is which carbonyl group reacts with methanol preferentially. On the basis of the material in [Section 17-6](#), the aldehyde group should be more reactive. Thus, as a working

hypothesis, propose a product with an acetal function derived from addition of methanol to the aldehyde carbon, and see if the NMR data agree.



One of the giveaways is the absence of a signal for a formyl (aldehydic) hydrogen in the δ 9.5–10 region of the spectrum. The mechanism of formation is precisely that outlined in [Section 17-7](#) of the textbook.

38. (c)

39. (b)

70. (d)

71. (b)

18

Enols, Enolates, and the Aldol Condensation: α,β - α,β -Unsaturated Aldehydes and Ketones

As the text continues to develop the chemistry of aldehydes and ketones, you will now see how the carbon adjacent to a carbonyl group can become nucleophilic. First, reactions of these new nucleophiles with common electrophiles like haloalkanes will be covered: alkylation reactions. More important are reactions of the nucleophilic α -carbons *α -carbons* of one carbonyl compound with electrophilic *carbonyl carbons* of another. They are generically termed **carbonyl condensation reactions**. You see them here for aldehydes and ketones: the [aldol condensation](#). (In a later chapter you will be introduced to the analogous reaction of carboxylic esters: the Claisen condensation.) The products of aldol condensations are α,β -unsaturated aldehydes and ketones, *α,β -unsaturated aldehydes and ketones*, which contain additional sites of electrophilic and potential nucleophilic character.

Pay close attention to the structural relationships between starting materials and products. These are the reactions of real-world organic synthesis, and they are among the most important ones you will see in this course.

Outline of the Chapter

18-1 The Acidity of α -Hydrogens: Enolate Ions

Making a nucleophile at the carbon α to a carbonyl group.

18-2 Keto-Enol Equilibria

18-3, 18-4 Halogenation and Alkylation at the α -Carbon

18-5, 18-6, 18-7 Aldol Condensation

A very important new reaction for synthesis.

18-8 through 18-11 α , β -Unsaturated Aldehydes and Ketones

The properties of the versatile products of aldol condensation.

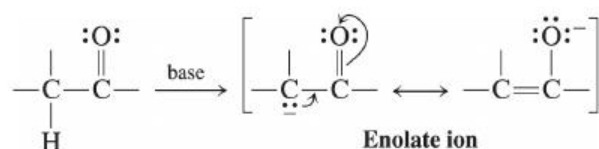
Keys to the Chapter

18-1. The Acidity of α -Hydrogens: α -Hydrogens : Enolate Ions

Hydrogens on carbons adjacent to carbonyl or related functional groups are, as a class, the most acidic of alkanelike hydrogens. A short list of representative pK_a values follows. Notice how acidity is enhanced by a second carbonyl group.

Compound	pK_a	Compound	pK_a
$\begin{array}{c} \text{O} \\ \parallel \\ \text{RCH}_2\text{COR}' \end{array}$	25	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{ROCCH}_2\text{COR} \end{array}$	13
$\begin{array}{c} \text{O} \\ \parallel \\ \text{RCH}_2\text{CR}' \end{array}$	20	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{RCCH}_2\text{COR}' \end{array}$	11
$\begin{array}{c} \text{O} \\ \parallel \\ \text{RCH}_2\text{CH} \end{array}$	17	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{RCCH}_2\text{CR} \end{array}$	9
$\begin{array}{c} \text{O} \\ \parallel \\ \text{RCH}_2\text{CCl} \end{array}$	16	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{HCCH}_2\text{CH} \end{array}$	5

Any of the boldface protons in the structures above may be removed by an appropriate base, thus giving rise to the central participant in the reactions of this chapter, the enolate ion.



The enolate is nucleophilic, capable of reaction with the typical electrophiles you've seen before. Their reaction with haloalkanes (alkylation) is the most general way to introduce alkyl substituents adjacent to carbonyl carbons.

18-2. Keto-Enol Equilibria

In this section more mechanistic detail is provided concerning the

reversible isomerization of vinyl alcohols (enols) to carbonyl compounds. This isomerization, although it favors the keto structure, allows small concentrations of enols to be in equilibrium with carbonyl compounds, and these enols can lead to productive chemical reactions. The process of “enolization” can take place in any carbonyl compound bearing an α hydrogen. α hydrogen. It may be catalyzed by either acids or bases.

18-3 and 18-4. Halogenation and Alkylation at the α -Carbon

The reactions in this section are mechanistically very similar to ones you know. The nucleophilic carbon of an enolate (or, in acid, an enol) attacks either a halogen or a haloalkane through an S_N2 pathway. Therefore, alkylations succeed best when the haloalkane is methyl or primary. Enamines can be used instead of enolates for alkylation and usually give superior results.

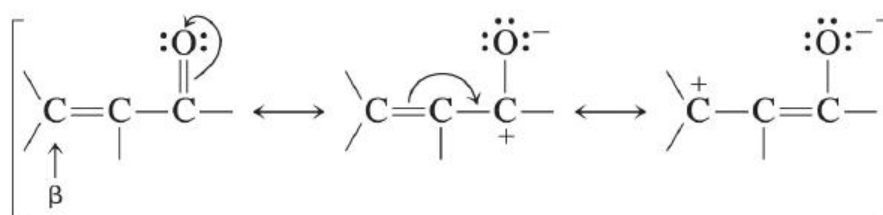
18-5, 18-6, and 18-7. Aldol Condensation

These sections cover the first, and probably the most important carbonyl condensation reaction, the aldol condensation. Its significance is due to its ability to make relatively complex structures, including rings, from much simpler starting molecules. It proceeds via addition of the deprotonated α -carbon of one carbonyl group to the carbonyl carbon of another. As you read this text section, notice the location of the new carbon–carbon bond in each aldol product. Cover up the starting materials and see if you can deduce their structures solely by looking at their aldol products. Although many aldol reactions between two identical molecules are useful, the most important applications fall into two categories: the so-called crossed aldols, in which two different reaction “partners” participate, and intramolecular condensations of dicarbonyl compounds, which make rings. Pay particular attention to the examples in this section, because they provide excellent illustrations for the various permutations of enolate and carbonyl “partners” that give

successful and useful results in aldol condensations.

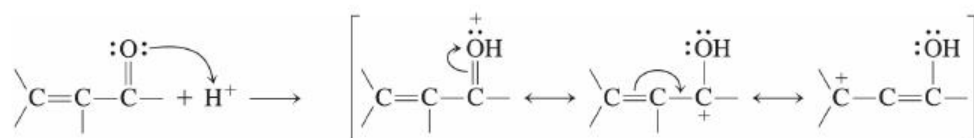
18-8 through 18-11. α,β -Unsaturated Aldehydes and Ketones

You see here logical extensions of carbonyl and alkene chemistry. The carbon-carbon double bond in an α,β -unsaturated carbonyl compound generally shows the same addition reactions typical of simple alkenes. However, the highly polar nature of the carbonyl group strongly affects the reactivity of the alkene functional group. This is best seen by the resonance forms in these compounds: Note in particular the presence of positive character at the β -carbon as well as at the carbonyl carbon.

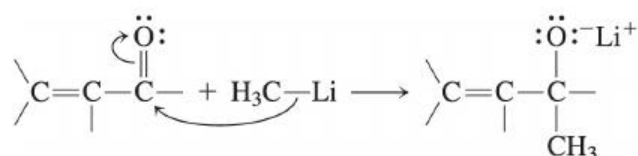


The wide range of *enone* chemistry is understandable in terms of just three fundamental mechanistic processes.

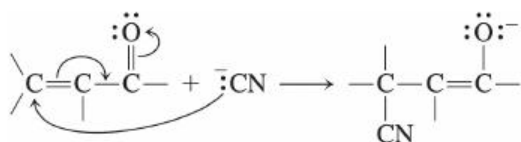
.. Electrophilic addition to carbonyl oxygen, e.g.,



!. Nucleophilic addition to carbonyl carbon, e.g.,

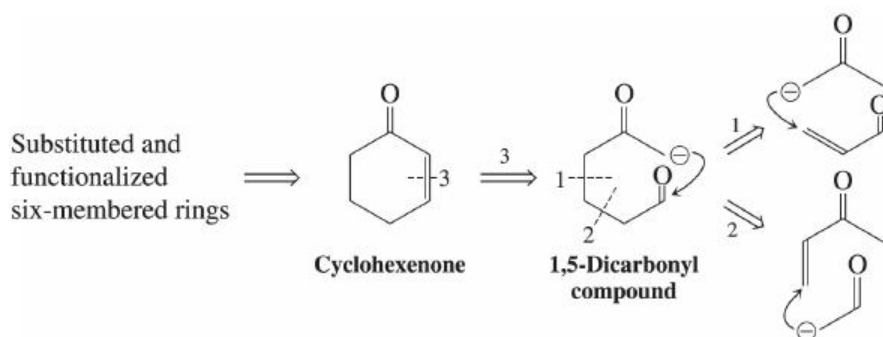


!. Nucleophilic addition to the β -carbon, β -carbon, e.g.,



Practice by writing mechanisms for examples in the text for which specific mechanisms are not pictured. Use the appropriate mechanism above to start, and follow it with the behavior expected for the product of that step.

Once you understand the mechanisms, concentrate on the synthetic applications of the process. Focus on the carbon-carbon bond-forming examples, with particular emphasis on the **Michael addition**, the 1,4-addition of enolates to enones or enals. The combination Michael addition-aldol condensation provides a powerful means of synthesis of six-membered rings, the **Robinson annulation**. Don't worry about all these people's names; just be sure to learn the retrosynthetic analysis for compounds containing six-membered rings.



The first step is Michael addition to give a 1,5-dicarbonyl compound. This, in principle, can be done in either of two ways (formation of either bond 1 or bond 2). Then intramolecular aldol condensation forms bond 3 of a cyclohexenone, which may be converted to a wide range of new compounds using the reactions of these sections.